MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS)
FIFTH SEMESTER
B.Sc(Hons) DEGREE EXAMINATION MATHEMATICS
(2024 ADMISION ONWARDS)
MG5DSEMAT303–EXPLORING THE HARMONY OF AUTOMATA

Duration: 2 hrs Maximum Marks: 70

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A
Answer any five questions. Each question carries 2 marks

1. Define a set. What is meant by the elements (members) of a set?[K][1].
2. Define a k-tuple and give an example of a 3-tuple. [U][1].
3. Give the formal definition of a finite automaton as a 5-tuple, naming each component. [K][2]
4. When does a nondeterministic finite automaton accept an input string? [U][2]
5. Define the language of a grammar, and define a context-free language. [K][3]
6. What is the role of the stack in a pushdown automaton? [U][3]
7. What are the three possible outcomes when a Turing machine is run on an input? [K][4]
8. Define a multitape Turing machine and write down its transition function. [U][4]

Part B
Answer any five questions. Each question carries 6 marks

9. What is a binary relation? Give an example of a binary relation on $\{1, 2, 3\}$. [U][1]

10. Define paths, simple paths, cycles, simple cycles, connected graphs and trees, illustrating each with a diagram. [U][1]

11. Prove that the class of regular languages is closed under the operation of concatenation, using a nondeterministic finite automaton in the construction. [A][2]

12. Give the formal definition of a regular expression and of the language $L(R)$ that it describes, explaining each of the six clauses of the inductive definition with an example. [U][2]

13. Prove the lemma that if a language is context free then some pushdown automaton recognizes it, describing carefully the construction of the PDA from a given context-free grammar. [A][3]

14. State and prove the pumping lemma for context-free languages.[An][3]

15. Define a nondeterministic Turing machine and describe its computation as a tree of configurations, explaining when the machine accepts and when it rejects. [U][4]

16. Prove that if an enumerator E enumerates a language A, then some Turing machine M recognizes A. [A][4]

Part C
Answer any three questions. Each question carries 10 marks

17. (a) Define the values TRUE and FALSE and the operations $\neg$ (NOT), $\wedge$ (AND), $\vee$ (OR), $\oplus$ (XOR), $\rightarrow$ (IMPLIES) and $\leftrightarrow$ (EQUALITY) by truth tables.

    (b) Verify the distributive law $P \wedge (Q \vee R) = (P \wedge Q) \vee (P \wedge R)$ and both De Morgan laws using truth tables.

    [U][1]

18. Prove that a language is regular if and only if some regular expression describes it. [An][2].

19. Prove that none of the following languages is regular. In each case exhibit the string chosen and examine every possible decomposition:

    (a) $\{0^n 1^n : n \geq 0\}$;

    (b) $\{ww : w \in \{0,1\}^*\}$;

    (c) $\{1^{n^2} : n \geq 0\}$.

    [An][2]

20. Prove that neither of the following languages is context free

    (a) $C = \{a^i b^j c^k : 0 \leq i \leq j \leq k\}$;

    (b) $D = \{ww : w \in \{0,1\}^*\}$.

    [An][3]

21.

    (a) Define string order on Σ^*.

    (b) Prove that a language is decidable if and only if some enumerator enumerates it in string order.

    (c) Explain why the corresponding statement without the ordering requirement characterises only the Turing-recognizable languages.

    [An][4]

22. State the Church–Turing thesis and discuss it in detail. [U][4]